

Unified Data Model and Knowledge Graph

Connecting software names and metadata to archived source

Andrew Nesbitt · ecosyste.ms

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Andrew Nesbitt

I build **ecosyste.ms**: APIs and datasets about packages, repositories and dependencies.

I work with **Alpha-Omega** on open-source security, and with **OpenAlex** and **UT Austin** on connecting research papers to the software they cite.

I'm a package management nerd and write about it at nesbitt.io.



Ecosyste.ms

Open metadata on software, dependencies, activity, security and funding.

52

Ecosystems

15.2M

Packages

185M

Package versions

25.3B

Dependency relationships

348M

Repositories

698K

Docker images & dependencies

37.4K

Security advisories

10.5M

Dependabot pull requests

46.9M

Issues

146M

Pull requests

2.3M

Maintainers

38M

Owners

\$48M

Open Collective funding

190K

GitHub sponsors

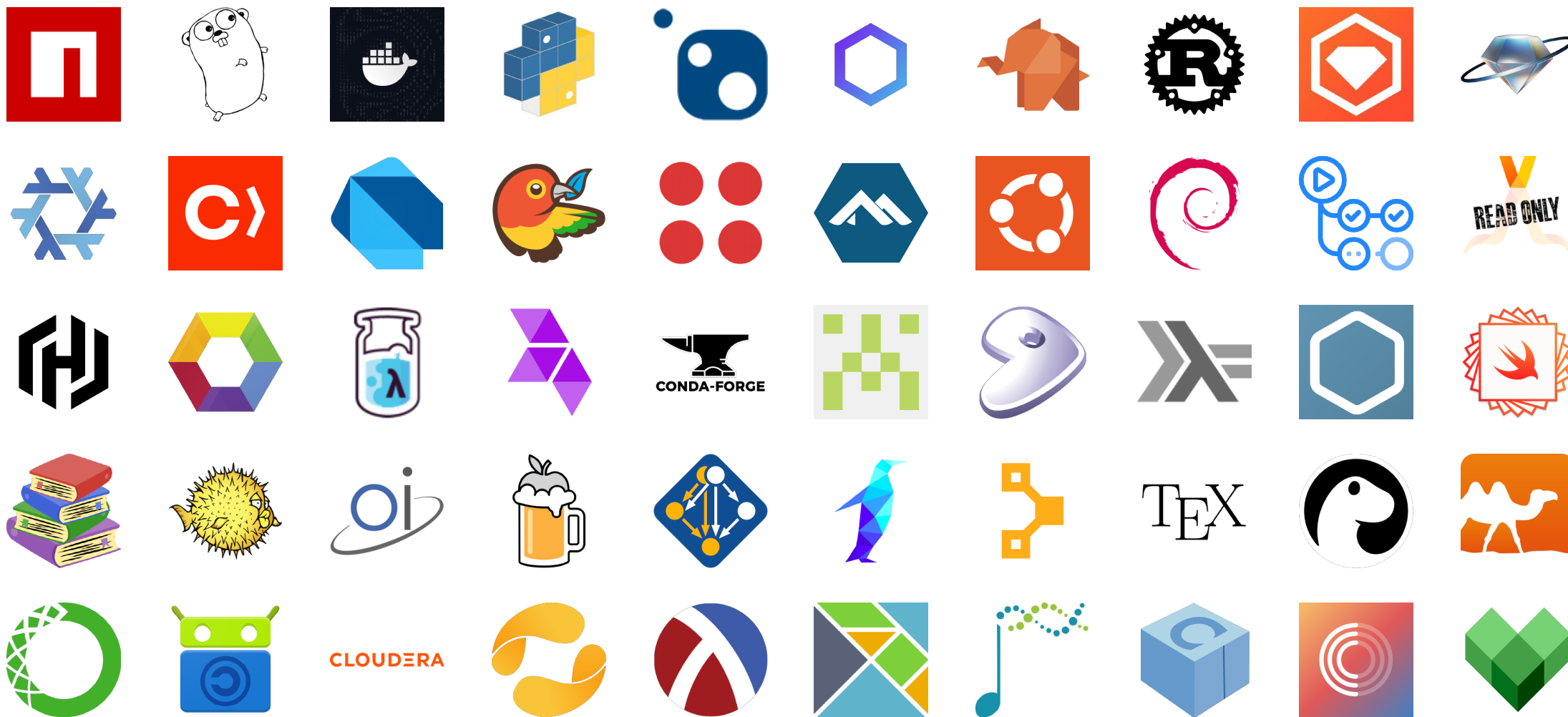
2M

Namespaces

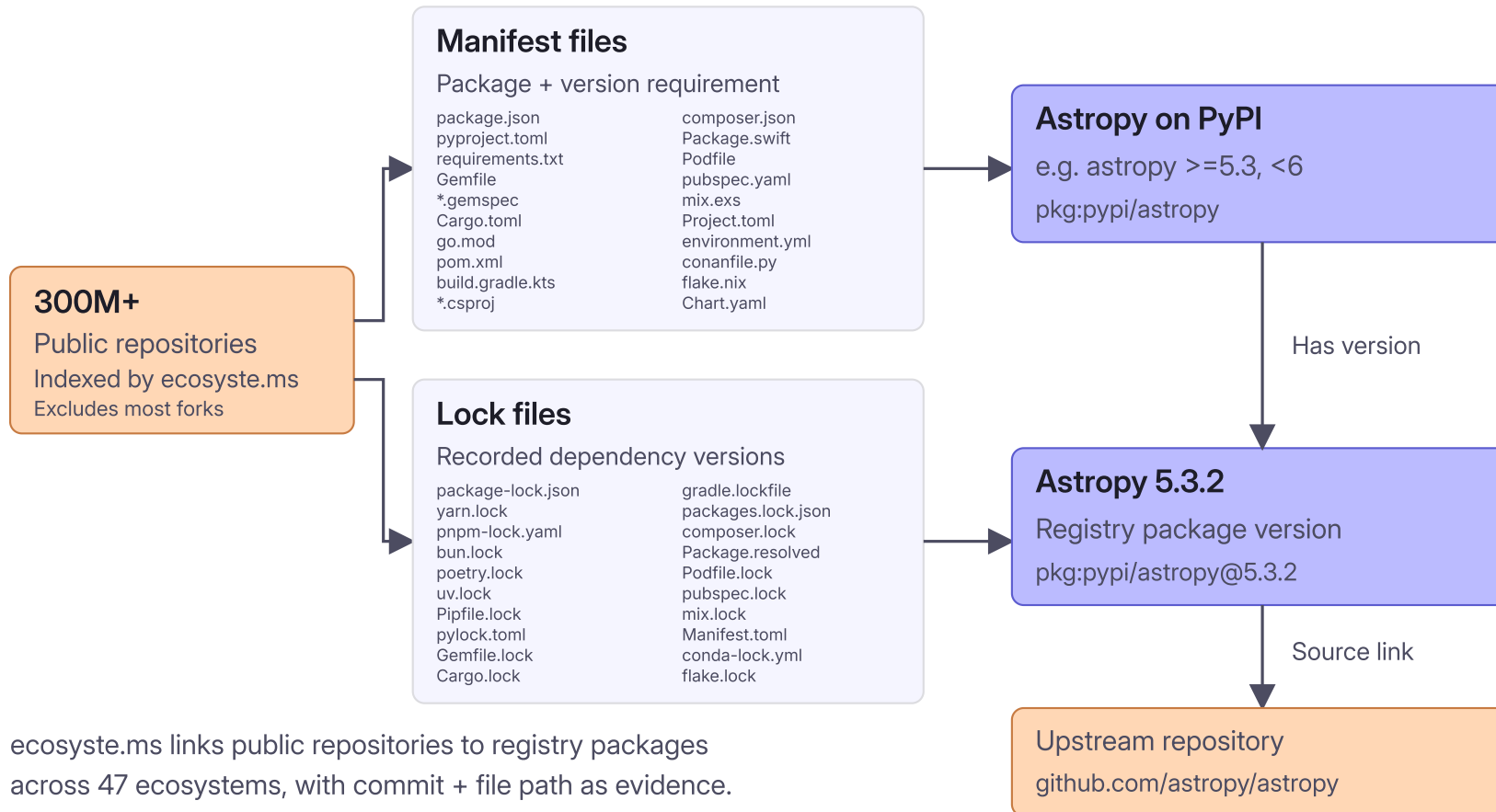
1.9B

API requests per month

Package registries



Dependencies



SBOMs

CycloneDX SBOM excerpt

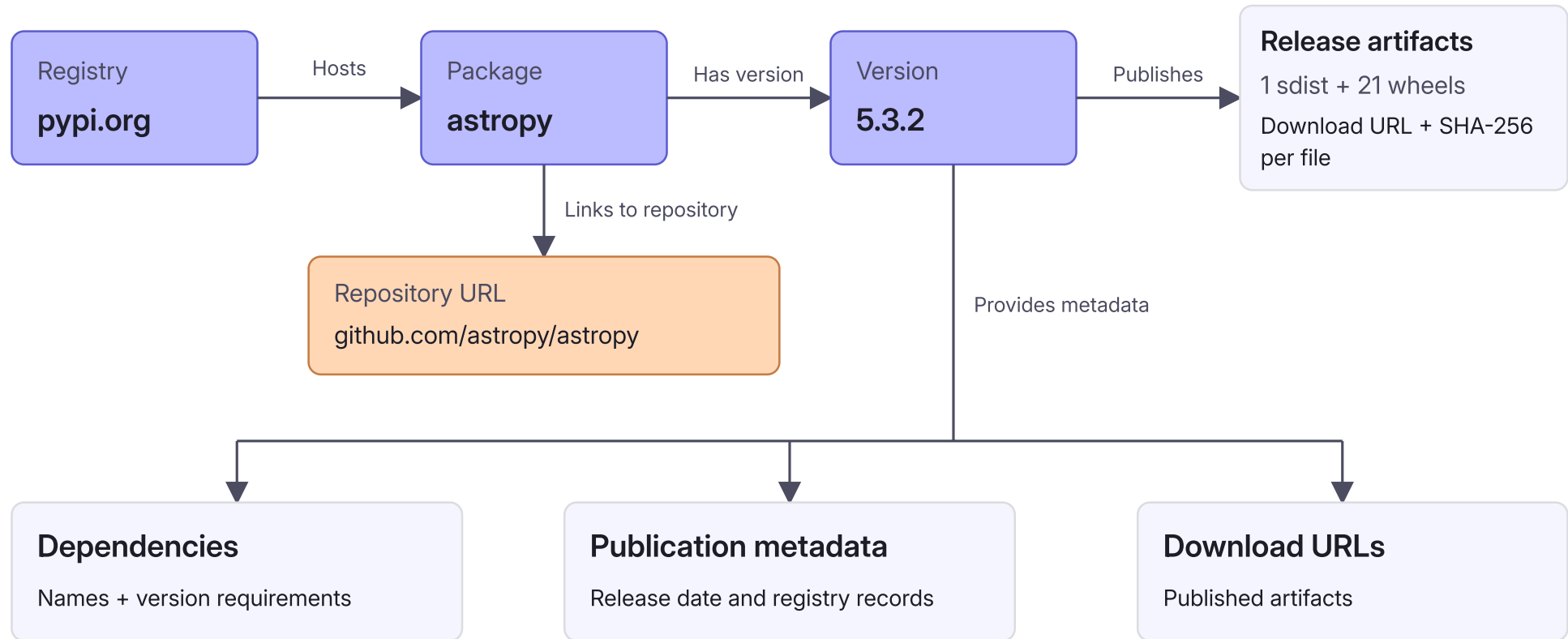
```
{
  "bomFormat": "CycloneDX",
  "specVersion": "1.6",
  "version": 1,
  "metadata": {
    "component": {
      "type": "application",
      "name": "astronomy-pipeline", "version": "1.0.0"
    }
  },
  "components": [
    {
      "bom-ref": "astropy",
      "type": "library",
      "name": "astropy",
      "version": "5.3.2",
      "purl": "pkg:pypi/astropy@5.3.2"
    },
    {
      "bom-ref": "numpy", "type": "library",
      "name": "numpy", "version": "1.24.4",
      "purl": "pkg:pypi/numpy@1.24.4"
    },
    {
      "bom-ref": "pyerfa", "type": "library",
      "name": "pyerfa", "version": "2.0.0.3",
      "purl": "pkg:pypi/pyerfa@2.0.0.3"
    }
  ],
  "dependencies": [
    {
      "ref": "astropy",
      "dependsOn": ["numpy", "pyerfa"]
    }
  ]
}
```

Astropy component

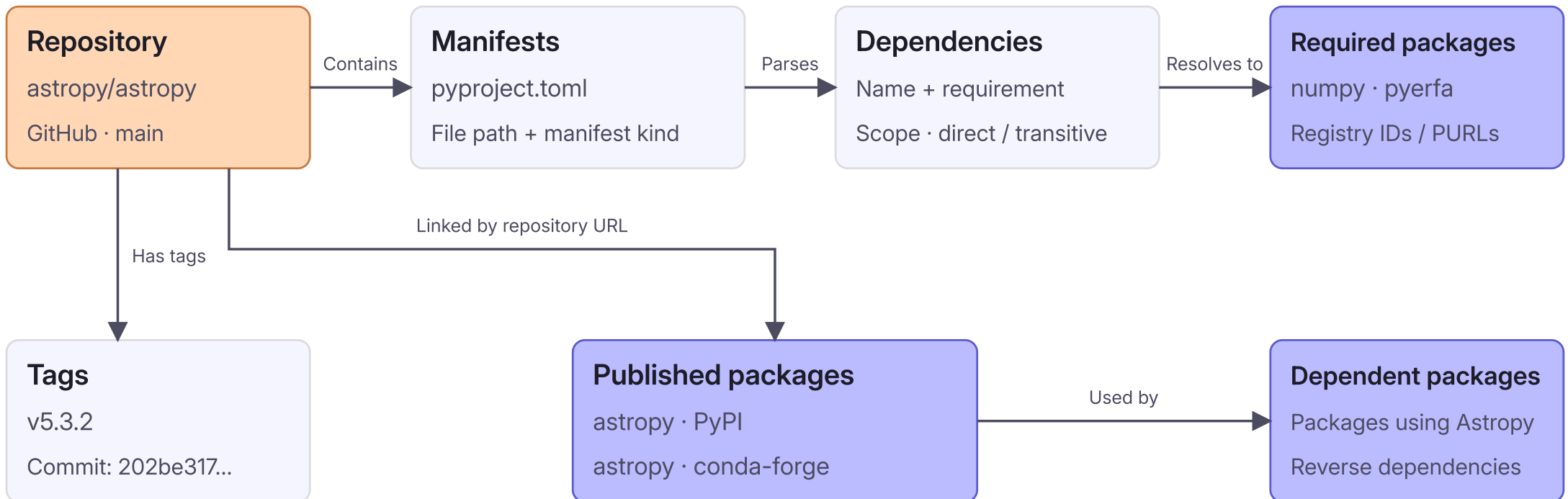
```
{
  "bom-ref": "astropy",
  "type": "library",
  "name": "astropy",
  "version": "5.3.2",
  "purl": "pkg:pypi/astropy@5.3.2"
}
```

ecosyste.ms answers millions of lookups
each day, adding license, security
and repository data to SBOMs.

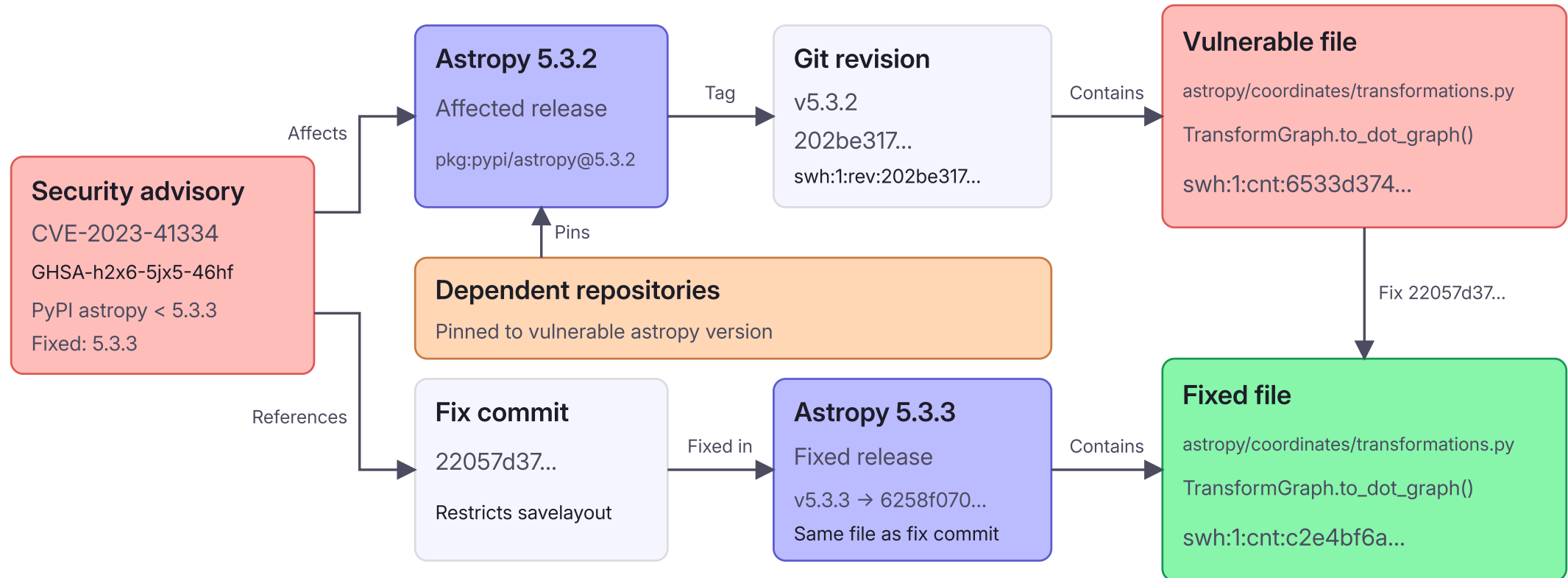
packages.ecosyste.ms



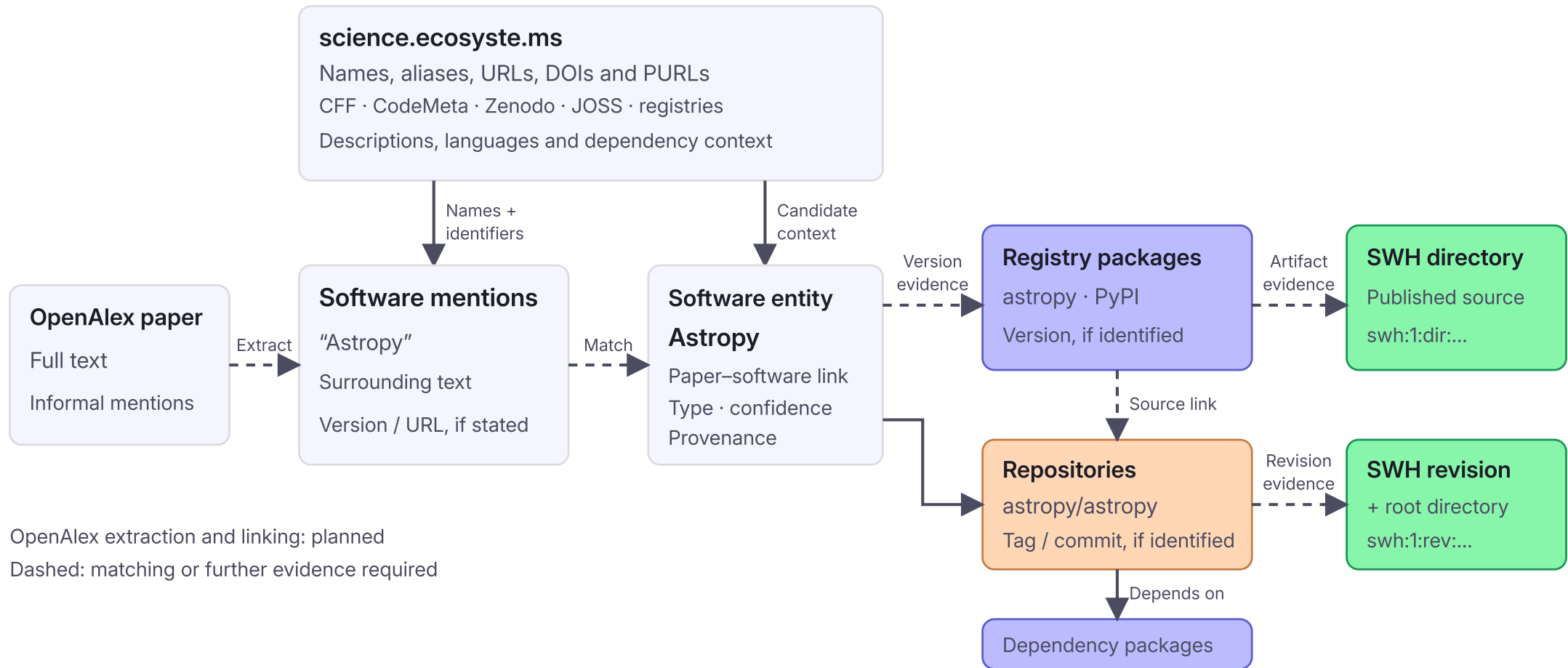
repos.ecosyste.ms



advisories.ecosyste.ms



science.ecosyste.ms



Human-meaningful

Recognise the name

Zooko's Triangle

https://en.wikipedia.org/wiki/Zooko's_triangle

Decentralized

No naming authority

Secure

Verify the binding

Human-meaningful

Recognise the name

Software mention

Astropy

Identity depends on context

purl

pkg:pypi/astropy@5.3.2

Binding: PyPI records

Zooko's Triangle

https://en.wikipedia.org/wiki/Zooko's_triangle

Decentralized

No naming authority

SWHID

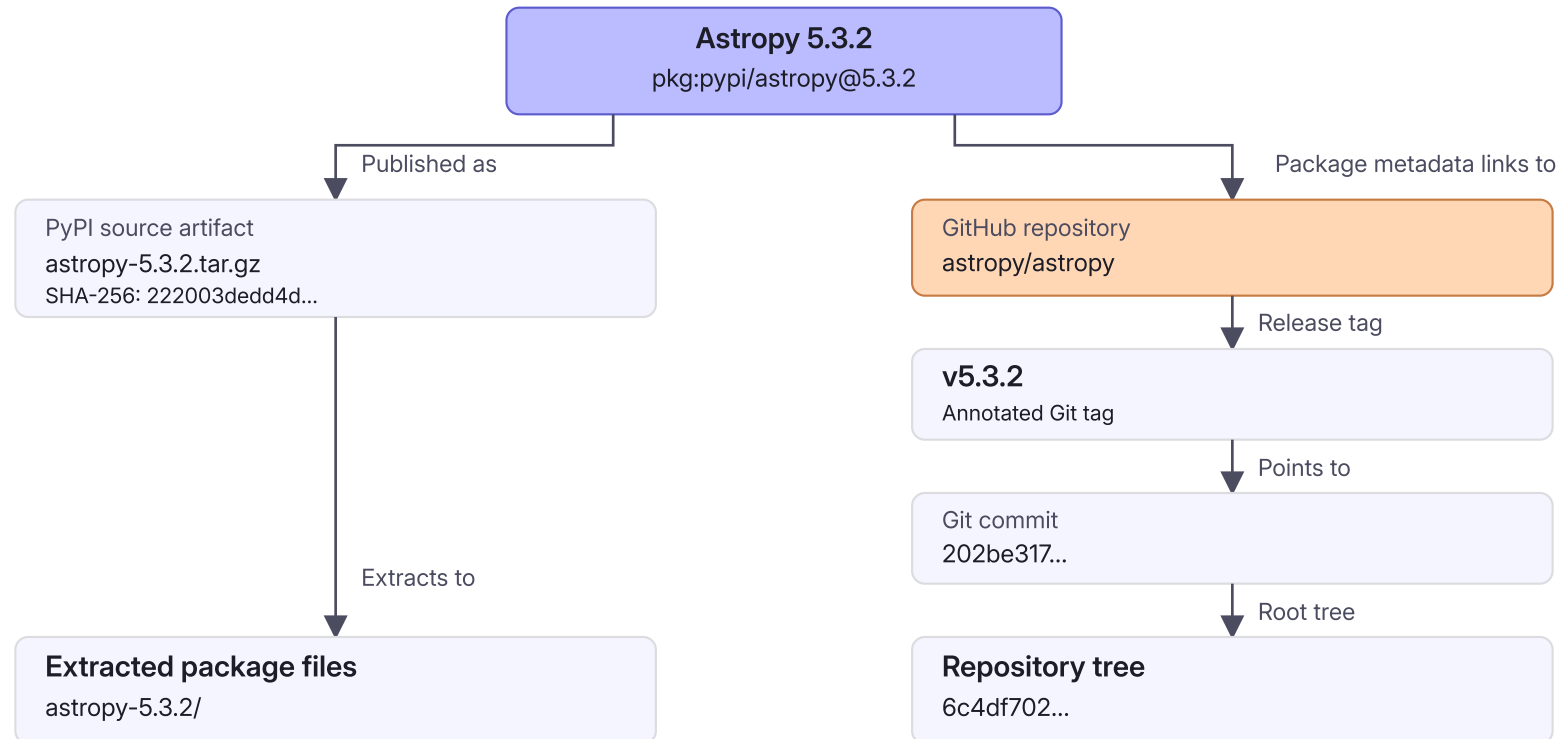
swh:1:dir:402de60d...

Computed from the source tree

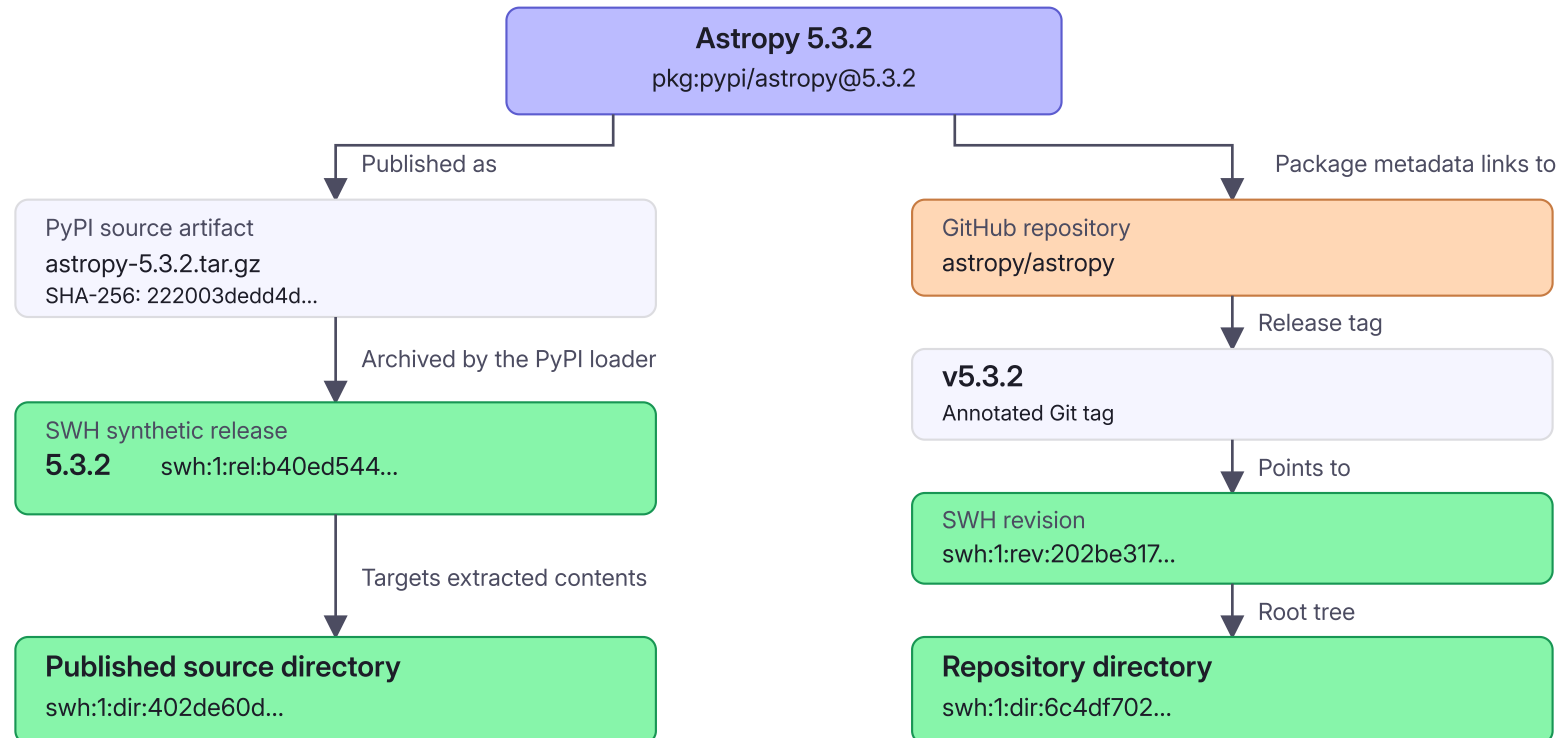
Secure

Verify the binding

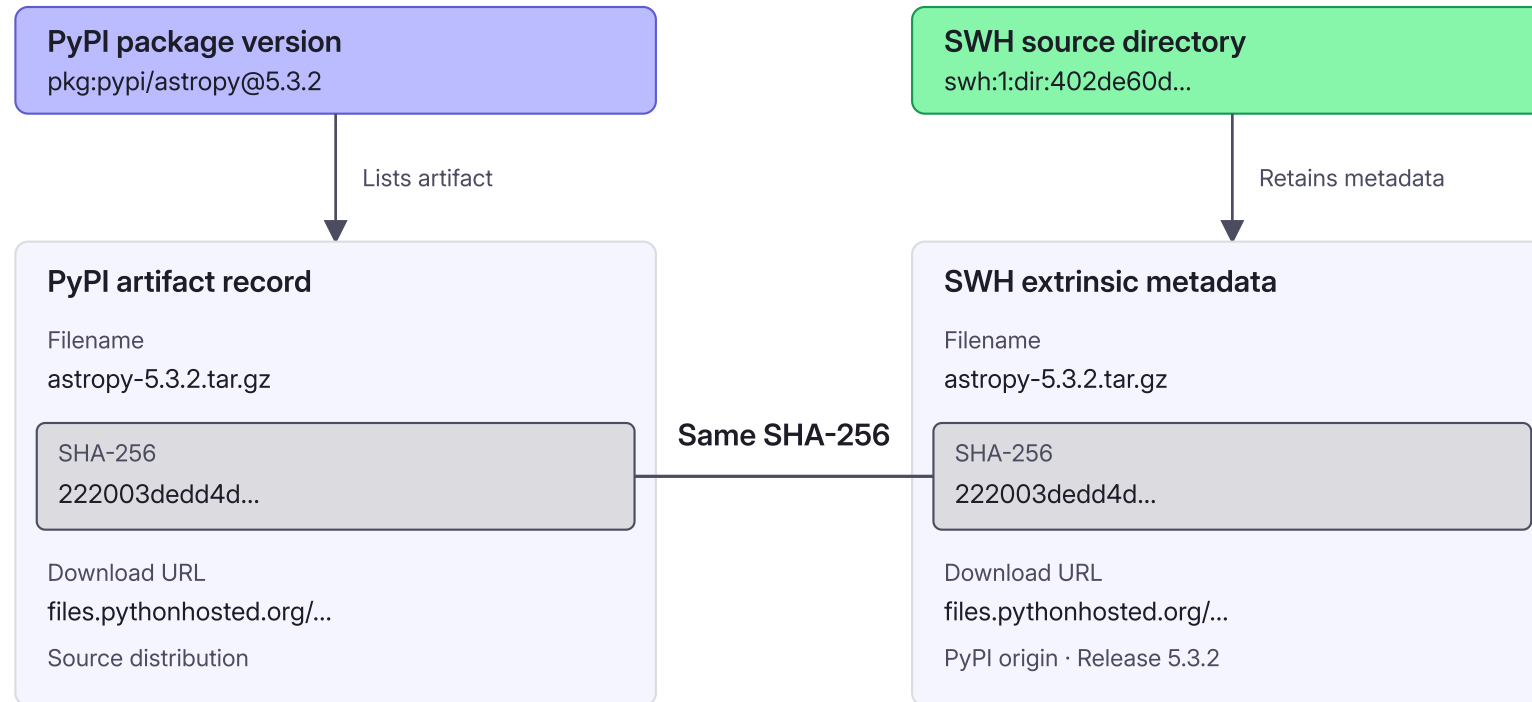
Package to Archive



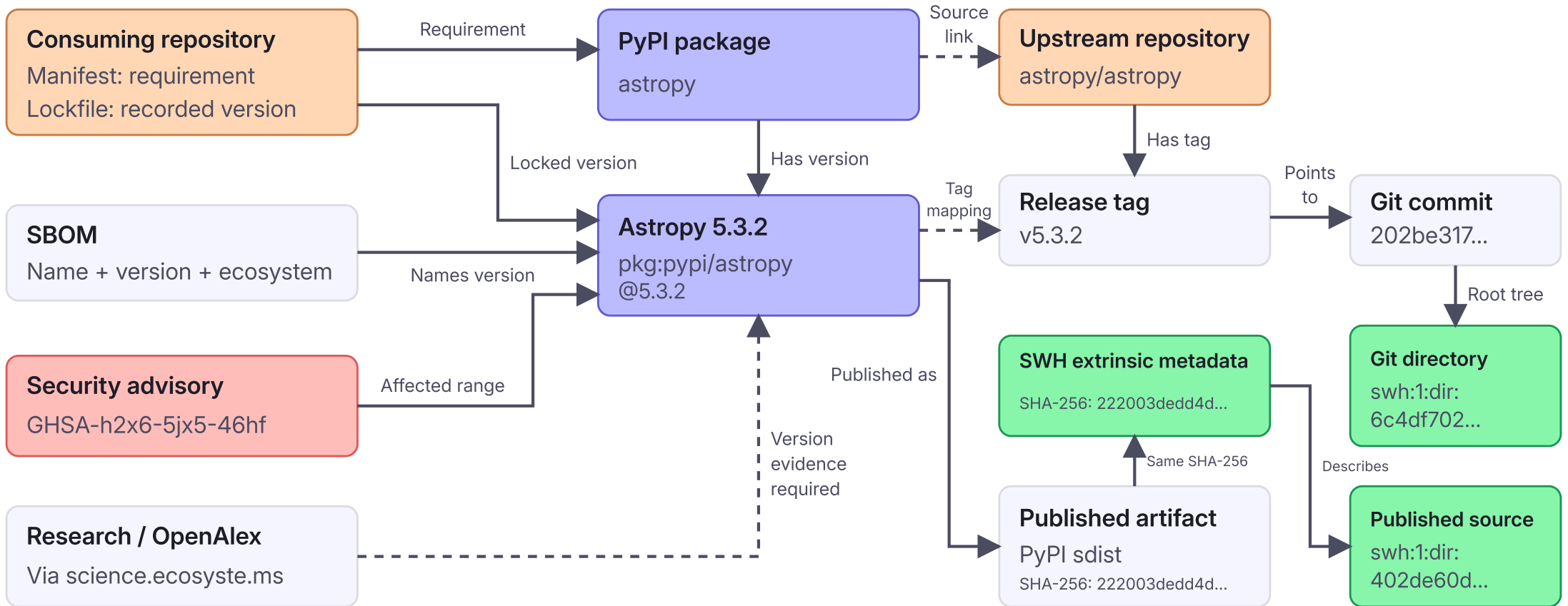
Package to Archive



Shared Content Addresses



Unified Data Model



Thank You · Q/A

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